

Multifloor Enterprise Network Design Report

1. Network Overview

This project implements a **multifloor enterprise network** for an **8-floor building**, supporting **8 departments**:

- Marketing
- Management
- Accounts
- Works
- HR
- Development
- Media
- Customer Care

Each floor operates as a **separate LAN**, connected through:

- Access switches on each floor
- Two multilayer (Layer-3) switches for core/distribution
- One building edge router
- Two ISP routers connected via serial links
- A remote **data-center (abroad)** connected using **Site-to-Site IPsec VPN**

The design ensures **secure communication, centralized services, scalability, and redundancy**.

2. IP Addressing & Subnetting Design

2.1 Internal Building Network

The entire building uses:

192.168.20.0/24

This network is subnetted to efficiently support different floor sizes.

2.2 Subnet Allocation Table

Floor / Dept	VLAN	Subnet	Mask	Usable Hosts	Gateway
Marketing	10	192.168.20.0/27	255.255.255.224	30	192.168.20.1
Customer Care	20	192.168.20.32/27	255.255.255.224	30	192.168.20.33
HR	30	192.168.20.64/27	255.255.255.224	30	192.168.20.65
Development	40	192.168.20.96/27	255.255.255.224	30	192.168.20.97
Management	50	192.168.20.128/27	255.255.255.224	30	192.168.20.129
Media	60	192.168.20.160/27	255.255.255.224	30	192.168.20.161
Accounts	70	192.168.20.192/27	255.255.255.224	30	192.168.20.193
Works	80	192.168.20.224/28	255.255.255.240	14	192.168.20.225

- Floors 1–7: /27 (up to 25 devices each)

- Floor 8: /28 (16 devices)

3. Server Network (Abroad / Data Center)

The remote data-center uses:

192.168.21.0/28

Server Allocation

Server Type	IP Address
Server Router	192.168.21.1
DHCP Server	192.168.21.5
DNS Server	192.168.21.6
Web Server	192.168.21.7
Email Server	192.168.21.8

3.1 Server-Side Protection (Access Restriction)

- Only **4 authorized servers** are connected
- No extra PCs or unauthorized devices are allowed
- Access is controlled using:
 - Port security
 - ACLs
 - VPN encryption

This ensures **data-center isolation and security**.

4. VLAN & Switching Design

Each department is assigned a dedicated VLAN.

VLAN ID	Department
10	Marketing
20	Customer Care
30	HR
40	Development
50	Management
60	Media
70	Accounts
80	Works
120	Voice

- **Trunk ports** carry multiple VLANs
- **Access ports** are department-specific

5. Layer-3 Switching & Inter-VLAN Routing

Multilayer switches provide **SVIs**:

- Each VLAN has a **default gateway**
- Enables **inter-VLAN communication**
- Reduces load on the router

5.1 DHCP Relay (ip helper-address)

All VLAN interfaces use:

```
ip helper-address 192.168.21.5
```

This forwards DHCP broadcasts to the **remote DHCP server**.

6. Network Services

6.1 DHCP

- Automatic IP allocation
- Centralized management

6.2 DNS

- Internal & external name resolution

6.3 Email & Web

- Inter-department communication
- Internal and external access

7. Building Automation Server (Works Department)

A **local automation server** controls:

- Lights
- Fans

- Electrical systems
- Network: 192.168.3.0/24
- Access limited to **Works department only**

8. WAN, NAT & ISP Connectivity

Two ISP routers are connected using **serial links**.

8.1 NAT Configuration

NAT is implemented using:

- **NAT inside** → internal VLAN interfaces
- **NAT outside** → ISP-facing interfaces
- **ACLs** define which internal IPs are translated

This allows:

- Internet access
- Secure VPN traffic handling
- Public IP conservation

9. Security Implementation

9.1 Device Security

- Console & enable passwords
- Encrypted passwords
- Warning banner (MOTD)

9.2 Secure Remote Access

- SSH v2 only

- Telnet disabled
- Local authentication

9.3 Access Control Lists

- VTY access restriction
- VPN interesting traffic definition

9.4 OSPF Dynamic Routing

- OSPF is configured with **Process ID 10**
- **Router ID manually assigned**
- Multiple networks advertised
- Uses **Area 0 (backbone)**

OSPF ensures:

- Automatic route learning
- Fast convergence
- Scalability across WAN and LAN

10. Site-to-Site IPsec VPN

A secure VPN connects:

192.168.20.0/24 <-----> 192.168.21.0/28

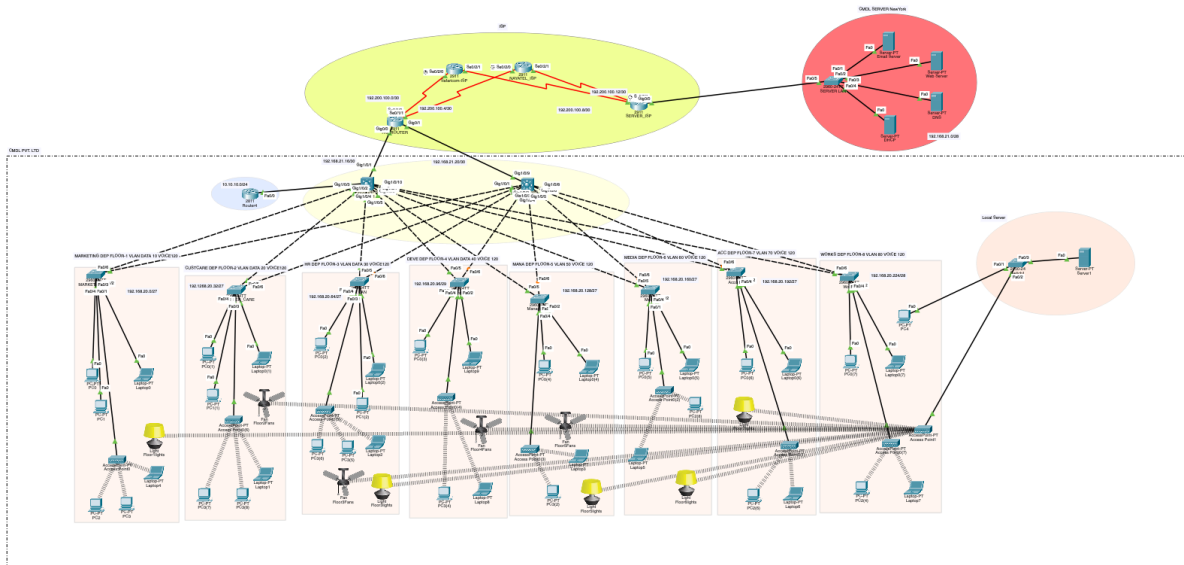
VPN Features

- AES-256 encryption
- Pre-shared key authentication
- ISAKMP Phase 1 & IPsec Phase 2

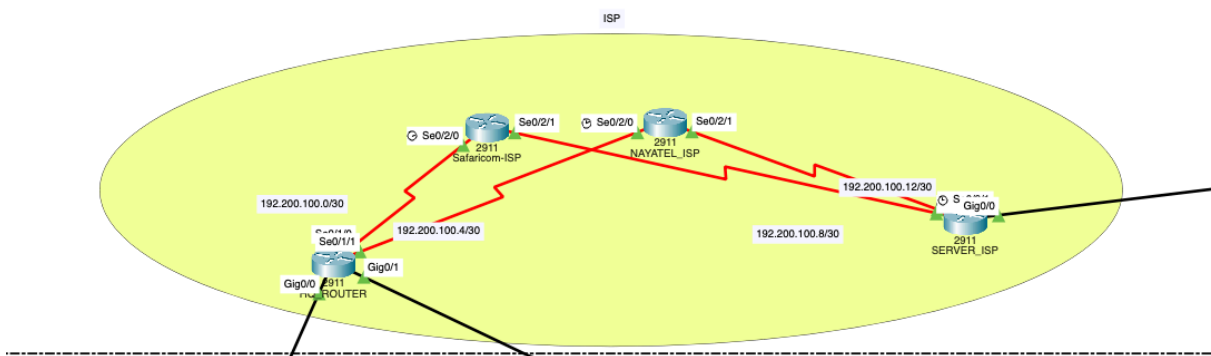
Provides:

- Confidentiality
- Integrity
- Authentication

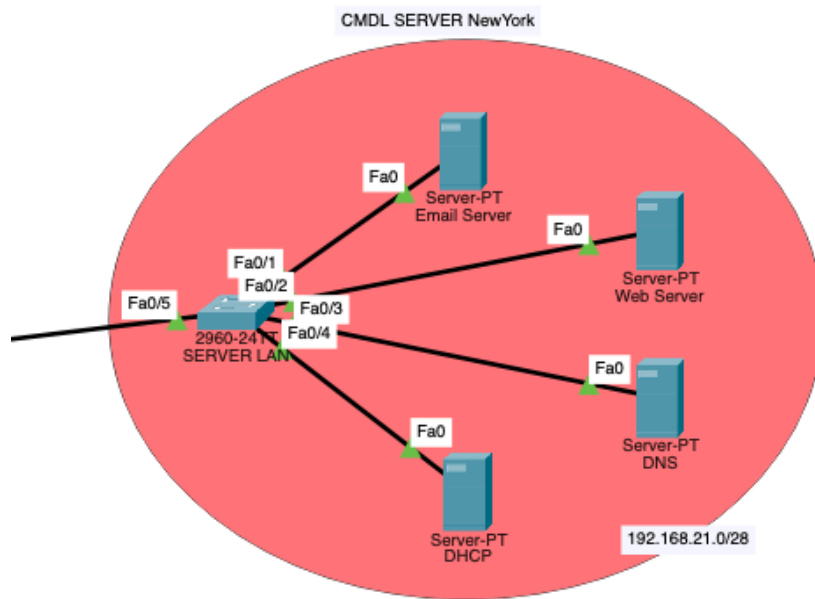
Topology



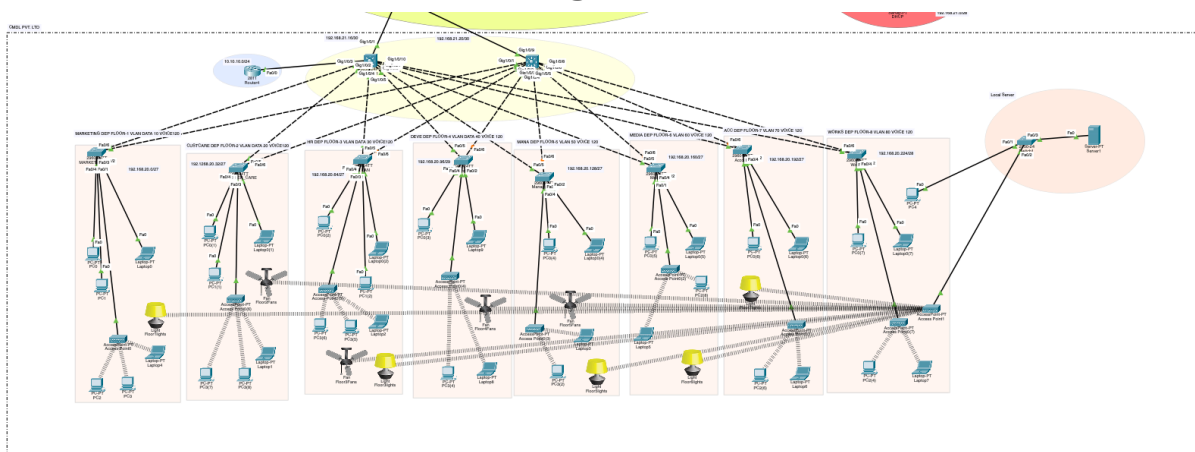
ISP



Server



Building Network



Building Local Server

VOICE 120

.20.224/28



Local Server

